

Subject Description Form

Subject Code	EEE502								
Subject Title	The Physics of Semiconductor								
Credit Value	3								
Level	5								
Pre-requisite/ Co-requisite/ Exclusion	Nil								
Objectives	This subject aims to let students gain insight in the physics and working principles of semiconductors. Transistors (circuits), LED (lighting), and solar cell (energy) are the underpinnings of modern technologies. The subject focuses on introducing the physics of these semiconductor devices, their historical origins, and how they can be optimized for future applications.								
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Understand the physics model of semiconductor structures and their significance in the electric properties of solids. b. Analyze the carrier statistics in semiconductors, carrier dynamics and the resulting conduction properties of semiconductors. c. Analyze semiconductor junctions at equilibrium and under bias, capacitance and current characteristics, and breakdown behavior. d. Describe and analyze MOSFETs, LEDs, and solar cells. e. Understand the basic semiconductor process steps.								
Subject Synopsis/ Indicative Syllabus	Introduction: Covers physical principles of semiconductors, including silicon and new materials, quantum effects, and band theory. Semiconductor blocks and devices: Overview of the basic building blocks (semiconductor junction, diode, etc.), MOSFETs, LEDs, solar cells, their operating principles and device equations. Semiconductor process technologies: Introduction of the major semiconductor processing steps, including crystal growth, photolithography, etching, diffusion, film deposition, etc. Trends of Semiconductor industry: Overview of the latest semiconductor technology trends, new materials, processes, as well as the future development of the semiconductor industry.								
Teaching/Learning Methodology	<table><tr><td>Lectures, supplemented with interactive questions and answers, and short quizzes</td><td>a, b, c, d, e</td><td>In lectures, students are introduced to the knowledge of the subject, and comprehension is strengthened with interactive Q&A and short quizzes</td></tr><tr><td>Tutorials, where problems are discussed and are given to students for them to solve</td><td>a, b, c, d, e</td><td>In tutorials, students apply what they have learnt in solving the problems given by the tutor.</td></tr></table>			Lectures, supplemented with interactive questions and answers, and short quizzes	a, b, c, d, e	In lectures, students are introduced to the knowledge of the subject, and comprehension is strengthened with interactive Q&A and short quizzes	Tutorials, where problems are discussed and are given to students for them to solve	a, b, c, d, e	In tutorials, students apply what they have learnt in solving the problems given by the tutor.
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	Assignments	a, b, c, d, e	Through working assignments, students will develop a firm understanding and comprehension of the knowledge taught.				
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed				
			a	b	c	d	e
	1. Assignments	40%	✓	✓	✓	✓	✓
	2. Examination	60%	✓	✓	✓	✓	✓
	Total	100 %					
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Assignments: Let students review the taught materials, do further reading for deeper learning, and understand better of the taught knowledge. Students may find these readings useful and will practice the obtained knowledge in the associated exercises. Examination: Tests will evaluate students’ understanding and knowledge learning from the subject.						
Student Study Effort Expected	Class contact:						
	▪ Lectures					30 Hrs.	
	▪ Tutorials					9 Hrs.	
	Other student study effort:						
	▪ Self-study					81 Hrs.	
	Total student study effort					120 Hrs.	
Reading List and References	Textbooks • Donald A. Neamen, Semiconductor Physics and Devices: Basic Principles, New York: McGraw-Hill, 2012.						
	References • S. M. Sze, Modern Semiconductor Device Physics, New York: Wiley, 1998. • J. A. Del Alamo, Integrated Microelectronic Devices: Physics and Modeling, Pearson, 2017.						